ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

Self Introduction: Written Script and One-Page Profile

Harshitha S

CSE Student | Aspiring AI Application and Agent Engineer

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Summary

I am a CSE student at St. Joseph's College of Engineering, interested in artificial intelligence and software development. My goal is to become an AI Application and Agent Engineer, building intelligent systems that solve practical, real-world problems.

I am developing foundations in programming, software engineering, and machine learning. I am exploring how AI models, tools, and data work together in useful applications. I value continuous learning, thoughtful problem-solving, and collaboration.

Introduction

Hello, I'm Harshitha S, a first-year CSE student at St. Joseph's College of Engineering. I chose computer science because I am curious about technology and how data and software can address real needs. I am interested in AI application and agent engineering, and I am building my software development and generative-AI skills.

I bring communication and leadership skills to my studies. I enjoy teamwork, sharing ideas, and taking responsibility. I want to deepen my technical knowledge, build practical projects, and contribute effectively with others.

Skills

Programming: Python and SQL; basic knowledge of C.
Professional: Communication, leadership, collaboration, and active learning.
Tools and interests: Canva, Microsoft PowerPoint, software development, AI applications, and intelligent agents.

Experience

Student, St. Joseph's College of Engineering | August 2026-present
Building programming and data-structures skills through coursework and independent learning. Developing experience through practice, project work, and collaboration.

Education

B.E. Computer Science and Engineering
St. Joseph's College of Engineering | 2026-2030
First-year student | Chennai, Tamil Nadu, India

Career Goal

My goal is to become an AI Application and Agent Engineer. I want to connect software, data, and AI models to build practical, reliable systems around people's needs.

Current Focus

Strengthen CSE fundamentals; practice Python, SQL, and C; learn machine-learning and generative-AI concepts; and build small projects. I also aim to keep improving communication, teamwork, and technical presentation skills.

Impressive Personality: Biographical Note

Jerry Liu | Co-founder and Chief Executive Officer of LlamaIndex | AI application builder | Technology entrepreneur

Introduction

Jerry Liu is a computer scientist and technology entrepreneur best known as the co-founder and CEO of LlamaIndex. The company creates software tools that help developers connect large language models (LLMs) with information in documents, databases, and other sources. That connection matters because a general-purpose model may not know an organization's private knowledge, recent updates, or specialized terminology. Liu's work focuses on the engineering layer that helps an AI application find and use relevant information when a person asks a question or delegates a task.

His career is a useful example of how a practical need can lead to a broader technology platform. Instead of treating a language model as a complete product, LlamaIndex builds the supporting components that let a model work with external data. This includes preparing information, organizing it for search, retrieving useful context, and passing that context to the model. The result is a more complete application in which the model's language ability is combined with information that is relevant to the user's situation.

Academic and Technical Background

Liu's public LinkedIn profile lists Princeton University. In a first-person account of LlamaIndex's early development, he describes following generative models through his academic and professional path, including research involving generative adversarial networks (GANs) and sensor compression. These areas are technically distinct, yet both involve representing and learning from complex data. That foundation helps explain the engineering perspective visible in his later work: AI systems need more than a model; they also need reliable ways to process inputs and connect them to useful outputs.

A strong technical foundation can support product thinking. When a model produces an answer, developers need to consider where its information came from, whether the information is current, and how the system should respond when evidence is missing. Liu's work highlights these practical questions and places data handling, retrieval, and software design alongside machine-learning capability.

The Problem That Started LlamaIndex

The project began with a concrete workplace challenge. Liu wanted to use a language model to prepare for customer meetings by drawing on details in internal tools and call notes. The model could generate fluent language, but it did not automatically have access to that company-specific context. Without a way to bring the relevant information into the interaction, the model could not reliably help with the task.

Early experiments explored how to connect outside data sources to language models. The project was first called GPT Index and later became LlamaIndex. Liu worked with Simon Suo to develop it into a broader open-source framework. The starting point was narrow and practical: make useful information available to a model at the right moment. The idea could then apply to many settings where people want to ask questions about their own documents and data.

Why the Context Layer Matters

A language model generates a response from patterns learned during training and the context supplied at the time of a request. For many workplace questions, the missing piece is not a more elaborate prompt but access to the right source material. A company policy, meeting note, product manual, or recent report may contain the details needed to answer accurately. Liu's early problem identified this gap between general language ability and local knowledge.

That insight shaped a useful way to think about AI products. The model is one component in a larger system that must locate trusted information, provide it in a usable form, and make the response understandable. This is especially important when information changes over time or differs between organizations. The application must connect to the correct source and handle it responsibly.

From an Open-Source Framework to a Developer Platform

As LlamaIndex grew, its focus expanded from an initial indexing framework into a collection of developer tools and services for building data-aware AI applications. The workflow commonly begins by loading information from files or other sources. The application then parses and organizes the material, creates indexes that support retrieval, and finds passages that relate to a user's question. A language model can use those passages as context when composing an answer.

This approach is often described as retrieval-augmented generation, or RAG. Retrieval gives the model relevant source material at answer time, while generation turns the question and retrieved context into a response. RAG can help connect a model to private or frequently updated information without requiring the model itself to contain every fact. Good results still depend on careful data preparation, retrieval quality, clear source handling, and evaluation. The framework addresses an important part of the application pipeline, while developers remain responsible for designing and validating the complete experience.

LlamaIndex's tools have covered data ingestion and parsing, retrieval, question answering over private data, and workflows that let agents work with documents and tools. These pieces reflect a layered view of AI application engineering. A useful system must manage the path from raw information to an action or answer, and each stage can affect accuracy, speed, and user trust.

Professional Career and Milestones

As co-founder and chief executive officer, Liu has helped guide the company's direction and communicate its tools to developers. LlamaIndex moved from an open-source project toward a wider developer platform as interest in large language models increased. In 2023, the company announced an $8.5 million seed round led by Greylock. This was a company milestone that supported its work on infrastructure for AI applications.

The company has continued exploring use cases that go beyond simple question answering. LlamaIndex announced LlamaCloud and LlamaParse in 2024 to address document processing and data extraction, and announced a spreadsheet agent in 2025 for tasks involving spreadsheet data. These product examples show how data access can develop into workflow support: a system may need to read a document, understand its structure, retrieve a detail, and help carry out a multi-step task.

Engineering Ideas Behind the Work

One important lesson from Liu's work is that AI application quality depends on system design. Developers have to decide how to divide documents into useful units, what metadata to preserve, how to search or rank results, and what evidence to provide to a model. They also need to test the system with realistic questions and examine incorrect or incomplete answers. These responsibilities connect software engineering with information retrieval, data management, and user experience.

The work also makes clear that agents need grounding and boundaries. An agent that can call tools or act on information needs suitable context, clear task steps, and ways to check its outputs. Connecting a model to data is therefore a foundation for more capable systems, but it does not remove the need for evaluation, safeguards, and thoughtful design.

Open-Source and Developer Community

Open-source software can help developers learn by inspecting working systems, trying examples, and contributing improvements. LlamaIndex's public project made its ideas accessible to a community experimenting with language-model applications. In June 2023, Liu reported that the project had reached 16,000 GitHub stars and 200,000 monthly downloads within six months. These are historical figures, but they indicate substantial early developer interest in tools for connecting LLMs with data.

Building Reliable Applications

The tools around a model also bring responsibilities. Developers need to monitor whether documents are parsed correctly, whether search returns relevant evidence, and whether the generated answer stays grounded in that evidence. They should consider access permissions when data is private and make it possible to review the material that informed an answer. If a system is uncertain, it should communicate that clearly instead of presenting a guess as fact. These design decisions help turn a promising prototype into a dependable product.

Impact on AI Application Development

Liu's work reflects a broader shift in AI development: the focus is moving from model capability alone toward complete applications that can use information and complete useful workflows. A model may be impressive in a demonstration, yet a real application must handle varied data, retrieve the right evidence, communicate uncertainty, and respond consistently. Tools for indexing and document processing help teams address these practical requirements.

LlamaIndex has contributed to the vocabulary and tooling of data-aware AI development by making common building blocks easier to explore. Developers can use these components to prototype systems that answer questions over private information, search collections of documents, or combine retrieved context with model reasoning. The broader value is the attention it brings to the application layer: data quality, retrieval, integration, and user needs are central parts of AI engineering.

Professional Qualities Reflected in His Work

A person's public career cannot establish every aspect of their private character. The qualities below are interpretations of Liu's visible work and product decisions. Curiosity appears in the exploration of how language models can use new and specialized information. Practical problem-solving is reflected in turning a meeting-preparation need into reusable software. Technical breadth is suggested by the combination of machine learning, data systems, retrieval, and application design.

Adaptability is visible in the way the project moved from indexing data toward document processing and agent workflows as developer needs changed. A builder mindset is reflected in creating tools that other developers can test and extend. These observations are based on public contributions, rather than assumptions about personal traits that cannot be verified from a professional profile.

Why Jerry Liu Is an Impressive Personality to Me

As a first-year CSE student at St. Joseph's College of Engineering who hopes to become an AI application and agent engineer, I find Liu's professional journey relevant to my goals. It begins with a specific user problem and develops into tools that can support many developers. This shows me that meaningful innovation does not always begin with inventing a new model; it can begin by noticing what people cannot currently do with the technology they have.

His work also encourages me to build a broad engineering foundation. To create reliable AI applications, I will need programming skills, data structures, databases, and an understanding of how information is retrieved and evaluated. I will also need to communicate clearly, collaborate with others, and listen to users so that a technical solution addresses a real need. These are skills I can develop through coursework, practice, and small projects.

The example of LlamaIndex motivates me to approach projects in stages: identify a concrete problem, understand the data involved, build a simple working version, test it with users or realistic examples, and improve it based on evidence. I want to learn how models, software tools, and trusted information can work together to create applications that are genuinely useful in everyday settings.

Key Takeaway

Jerry Liu's journey illustrates curiosity, practical problem-solving, technical depth, and adaptability through the development of tools for data-aware AI. The lesson I take from his work is to look beyond the model and design the complete system around it: the data, retrieval, evaluation, tools, and user experience. By combining strong fundamentals with consistent project work, I hope to grow into an engineer who builds responsible AI applications and agents that solve meaningful problems.

Sources and Further Reading

Prepared as an educational profile based on the supplied Jerry Liu college profile and its references: Jerry Liu's public LinkedIn profile; "Building the data framework for LLMs," LlamaIndex (2023); the Latent Space interview "RAG Is A Hack"; LlamaIndex materials on LlamaCloud and LlamaParse (2024); the LlamaIndex Spreadsheet Agent announcement (2025); and LlamaIndex/Weaviate materials on retrieval-augmented generation. Professional qualities are interpretations of published work, not claims about private character.

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

Week 1: Course Introduction and Growth Mindset

The course opens with two preview lectures that recap an earlier course, “Developing Soft Skills”. That course dealt with perception, self-management, excellence, conflict, stress, habits, listening, telephone skills, technology, communication barriers, non-verbal communication, presentations, reading and trust. The big new topic here is mindset: a person’s habitual attitudes and ways of reading a situation.

In the shoe-salesman story, nobody wears shoes. Is that “no chance” or a “splendid opportunity”? The same situation draws opposite responses. Drawing on Carol Dweck, the lectures set a fixed mindset against a growth mindset. People with a fixed mindset avoid challenges, resist change, dwell on their limits, reject criticism and stop learning once school is over. Growth-mindset people welcome challenges, learn from failure and treat learning as lifelong.

Learning mindsets form early. A belief like “I’m not smart enough” limits a person, but it isn’t a fact about their ability. The suggested “secrets” are to behave like a baby (open and unafraid), put up with critical people, reject negative stereotypes and stereotype threat, and look for a sense of belonging. The Hawthorne effect is offered as evidence that belonging and attention motivate people more than incentives do.

Week 2: Time Management, Procrastination and Assertiveness

Time can’t really be managed. What we manage is our habits and our perception of it. Monochronic people stick to rigid schedules, while polychronic people stay flexible. Others tend to judge our personality by our punctuality and by how we behave under deadline pressure. To control time, log it, look for “time leakage” (interruptions, indecision, poor organisation, social media, unmanaged email) and make room for productive work. Working at your biological peak time helps. Covey’s urgent/important quadrants and the 80/20 Pareto principle help with prioritising, and “time stealers” such as waiting and travel can be turned into “time gifters.”

The lectures trace procrastination to Parkinson’s law, fear, laziness, perfectionism and addiction, along with the plain feeling of being overwhelmed. The remedies are:

- Eat the elephant bite by bite.
- Do the “frog” (the hardest task) first.
- Use the Kaizen one-minute principle.
- Keep a clean workspace.
- Set goals you actually want and declare them publicly.
- Reward yourself.
- Use the Zeigarnik effect: once you start, you feel compelled to finish.

The week ends with a first look at assertiveness, or not saying “yes” when you mean “no.”

Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive people on body language, listening, feelings and outcomes. Assertiveness means stating your rights without violating anyone else's, and it ideally ends in a win-win. Practical steps include checking your relationships, dropping self-debilitating thoughts, avoiding discouraging people, following Stanley and Nancy's commandments and learning four ways to say no.

On anger, Aristotle's point is that the hard part is being angry at the right person, to the right degree, at the right time. Techniques include postponing your reaction, counting, deep breathing, physical exertion, Lincoln's unsent letter, silence and channelling anger into creative work. The lectures then turn to positive thinking through the story of two hospital patients and a "window" that faced a blank wall. Changing negative thinking takes desire and determination, introspection and discarding limitations. It also means building a positive self-image through repetition and visualisation, and bringing body, mind and world into sync.

Week 4: People Skills and Being Attractive

Forty traits make others dislike you, and their opposites make people like you. Examples include egotism, aggression, pessimism, fault-finding, gossiping, false promises, moodiness and favouritism, set against selflessness, optimism, giving credit, politeness, trustworthiness, honesty and humility. For dealing with gossip, Socrates' three-filter test (truth, goodness, utility) is suggested.

Attractiveness has physical, mental, emotional and spiritual layers. Physically, it means neat footwear, a fit body, good diet and sleep, and a signature personality. Mentally, it means intellectual charm and a touch of mystery. Emotionally, it means remembering names, valuing yourself and being yourself. Spiritually, it comes down to mindfulness, deeper connection, helping without expectation and "paying it forward."

Week 5: English Skills and Common Errors

English is the global lingua franca, and avoiding common errors is presented as part of soft skills. Using Sonia Sharma's Common Errors in English, the lectures work through repeated self-scored exercises and explanations. Topics include confused word pairs (resume/résumé, principal/principle, alone/lonely), Indianisms (passed out, joint family, cousin brother) and subject-verb agreement (collective, uncountable and plural-form nouns). The lectures also cover prepositions (since/for, at/in, by/on), articles, collocations and word choices like weather/climate. Twain and Branson are quoted on learning by exposure rather than by memorising rules.

Week 6: Humour and Money

Humour reduces stress, improves health and helps communication. It works at three levels: a smile, hearty laughter, and sarcasm, which is best avoided. Adults laugh far less than children (about 4 times a day versus 300). At work, humour resolves conflict, boosts creativity and productivity, and lets you speak the truth lightly. Suggestions include a joke board, a humour club and a humour hall.

The focus then turns to money and personality. How you treat rich and poor people, how you borrow and lend, and how you handle trust all reveal emotional maturity and integrity. Wealth-building advice includes tracking income and expenses, separating need from greed, avoiding impulse buys, and becoming indispensable through devoted, excellent work, with Michelangelo, Martin Luther King and Lincoln as examples. Real wealth also includes things money can't buy.

Week 7: Health and Love

Health affects personality, first impressions and employability, since healthy people are seen as disciplined and reliable. Managing it rests on:

- Regular exercise (morning is best, with tips for sticking to it).
- Diet: regular meals, a heavy breakfast and a light early dinner, fruits and vegetables, about 2 litres of water, and no white foods or soft drinks.
- Sleep: 7-8 hours, a good sleep environment, and avoiding gadgets, late heavy meals and alarms.

Love is defined broadly, drawing on Shakespeare, D. H. Lawrence, Gibran, Lovelace and Corinthians: real love is constant, free, patient and unconditional. Managing love means reading body language and timing, letting relationships mature through stages, handling rejection gracefully and keeping romance separate from work.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette make up character (“when character is lost, everything is lost”). Professional ethics combine written codes with society’s moral expectations. Poor etiquette and business etiquette are covered in detail: introductions, not interrupting, confidentiality, not criticising your company, and praising in public but reprimanding in private.

The mind is powerful but scattered, and it can be trained through auto-suggestion, positive self-talk and effective thinking (gather facts, sleep on it, trust your gut). Memory techniques include association through absurd images, remembering names and faces, number-image links, memory aids and mental exercise. The final lecture urges care for the environment, with practical steps (save water and electricity, avoid plastic, use public transport, plant trees) and a shift from anthropocentric to eco-centric thinking. It closes with Gandhi’s line that the earth provides enough for everyone’s need but not everyone’s greed.

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis

Strategic Career Roadmap: AI Application and Agent Engineer



Target role | AI Application and Agent Engineer

Harshitha S | First-Year CSE | St. Joseph's College of Engineering | 2026-2030

Goal: Build practical, reliable AI applications and agent-based systems through strong fundamentals, projects, evaluation, and team experience.

Executive Summary & Career Vision

1 Vision

Build the technical and professional capability to create useful, reliable AI applications and agent systems that solve real problems.

2 Starting Point

First-year CSE student with Python and SQL programming skills, basic C knowledge, communication and leadership skills, and an active interest in learning.

3 Roadmap

Strengthen CSE fundamentals, learn AI application development, build and evaluate projects, gain team and industry experience, and prepare for internships.

This roadmap is a development plan for the four-year degree, not a claim of current professional experience.

SWOC Matrix at a Glance

SWOC Matrix at a Glance

S Strengths

Python and SQL foundation; basic C knowledge; communication and leadership; active learner with an early career focus.

W Weaknesses to Develop

AI/ML and LLM concepts are developing; need project evidence and more practice with APIs, Git, testing, and deployment.

O Opportunities

Four-year academic runway; campus projects; open-source learning, clubs, hackathons, and internships; combine CSE and AI skills.

C Challenges

Fast-changing tools; competitive roles; reliability, privacy, and security requirements; balance fundamentals, projects, and coursework.

Strengths

1. **Technical foundation**
   Python and SQL support scripting, data handling, and software projects. Introductory C reinforces core programming concepts.

2. **Communication and leadership**
   Explain technical work, coordinate tasks, listen to feedback, and contribute responsibly to group projects.

3. **Curiosity and initiative**
   Enjoy learning new things and have selected a career direction early, leaving time to explore and build experience.

4. **Collaborative mindset**
   Work with peers and users, share ideas, and learn through projects and constructive feedback.

Weaknesses and Opportunities

Weaknesses and Development Areas

AI foundations

Build knowledge of data, probability, machine learning, and language-model concepts before relying on frameworks.

Applied experience

Turn learning into projects that document the problem, design, implementation, tests, limitations, and lessons.

Software delivery skills

Practice Git, APIs, databases, debugging, testing, and deployment alongside AI skills.

Early-career exposure

As a first-year student, gain experience steadily through coursework, small projects, faculty guidance, and peer collaboration.

Opportunities

Academic runway

Use each year of the degree to build progressively deeper skills instead of trying every AI tool at once.

Project-led learning

Start with campus needs such as a canteen seat estimator or college FAQ assistant; define success and test with users.

Learning communities

Use documentation, faculty guidance, coding clubs, open-source work, workshops, and hackathons to learn with feedback.

Interdisciplinary work

Explore AI in education, accessibility, health, finance, and campus services; domain understanding makes projects more meaningful.

Professional exposure

As skills grow, seek collaboration, research opportunities, mentorship, and internships. Treat these as learning opportunities, supported by working project evidence.

Challenges & Action Plan

Challenges

Fast-changing tools

Frameworks evolve quickly. Learn underlying ideas and keep project dependencies and documentation current.

Proving capability

A career goal alone is not evidence. Build working software with readable code, tests, and thoughtful trade-offs.

Trustworthy AI

Evaluate outputs, protect data, show sources where possible, and keep human review for consequential actions.

Managing breadth

Balance CSE fundamentals, AI, projects, and coursework with a focused weekly routine and a few learning goals.

Action Plan: Closing the Gaps

1 AI Foundations

Study programming, probability, data preparation, ML basics, LLM concepts, and responsible use. Explain ideas in short notes.

2 Project Portfolio

Build two small programming or data projects, then an AI application with a defined user problem and documented evaluation.

3 Software & Agent Skills

Practice Git, APIs, databases, testing, and deployment. Add tool use and multi-step workflows after a basic application works; include safeguards.

Year 1-2: foundations and projects | Years 2-3: agent workflows | Years 2-4: collaboration and internships

Action Plan: Using Opportunities

Action Plan: Using Opportunities

Build useful solutions

Choose a real need, speak with intended users, define success, and build a simple first version.

Show evidence of learning

For each project include a README, setup steps, demo, tests, limitations, and next improvements.

Learn in public responsibly

Share progress and technical lessons. Never post private data, credentials, or claims beyond the work.

Grow through collaboration

Join a coding or AI community, contribute a small open-source improvement, or take part in a team challenge.

4-Year Milestones Matrix

Year	Primary focus	Target deliverable	Readiness evidence
Year 1 2026-27	CSE fundamentals: Python, SQL, C, Git, problem-solving.	Two small projects; clear README; regular practice.	Explain code, use version control, show tests.
Year 2 2027-28	ML basics, APIs, databases, backend, LLM application patterns.	Deploy an AI app; evaluate examples and record failures.	Connect model to data; discuss quality, cost, privacy.
Year 3 2028-29	Agent workflows, tools, retrieval, evaluation, cloud, security.	Bounded agent project with safeguards and user feedback.	Handle errors, permissions, and human review.
Year 4 2029-30	Reliability, deployment, specialization, career preparation.	Capstone, polished portfolio, internship if available.	Present end to end and explain trade-offs.

Milestone dates are estimates for a degree starting in 2026; adjust to the official calendar and progress.

Conclusion & Key Takeaways

Conclusion & Key Takeaways

1. **Think first, then build**

   Learning before coding can help me choose problems and methods that match real needs.

2. **Fundamentals plus AI**

   A strong CSE foundation and carefully chosen AI projects can support meaningful development.

3. **Verdict**

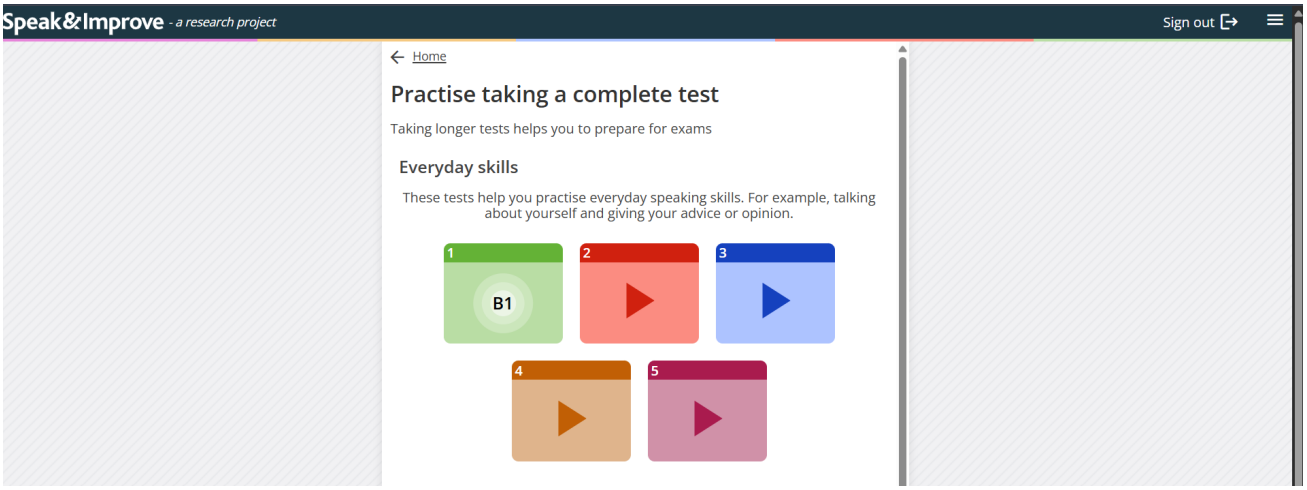
   With discipline, feedback, and steady practice, I can work toward building practical and responsible AI applications and agents.

Career direction

By developing step by step, I can work toward becoming an AI application and agent engineer who builds practical, reliable systems. My next priority is to strengthen fundamentals and demonstrate them through projects, evaluation, and teamwork.

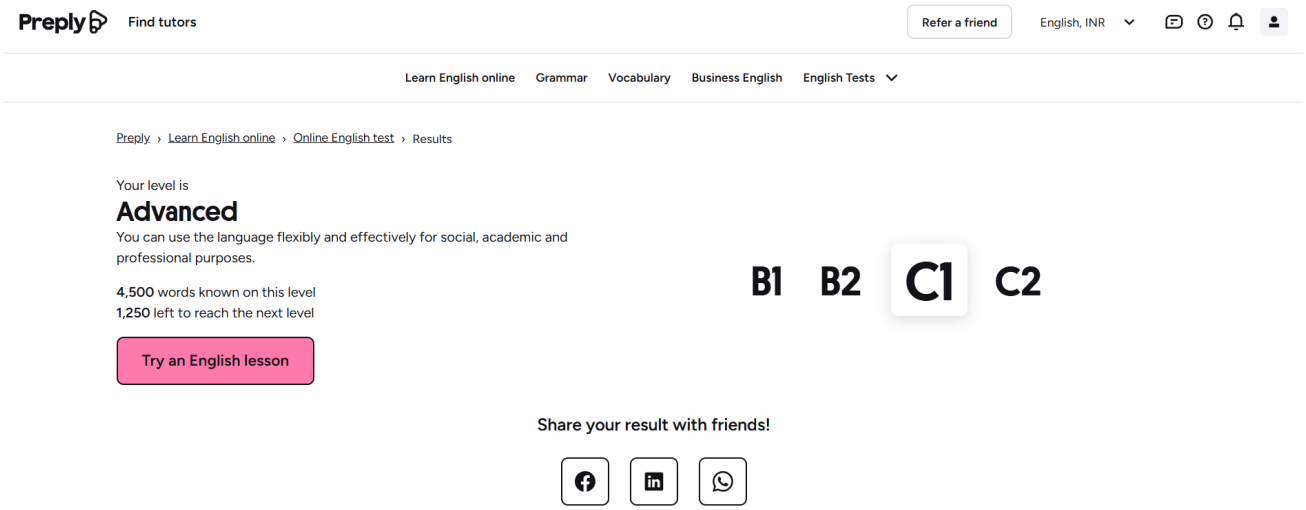
Write and Improve / Speak and Improve

Speak and Improve: score sheet (Task: Browsing on your phone)



Write and Improve / Speak and Improve

CEFR score card



My CEFR level is C1(advanced).

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must
5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing
11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:** The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   Answer (Screen Size): A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".)

   (b) **Answer:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully

10. (a) **Answer:** can

    (b) **Answer:** must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for every account.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.
5. Check that a website address begins with "https" and shows a padlock before entering any sensitive information.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or the National Cyber Crime Helpline (1930).

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Selected Question: 27.a

Bharatanatyam: The Dance That Carries Tamil Nadu's Soul

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student's formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

Question 28 (b) Formal letter of request to the Principal

Selected Question: 28.b

Harshitha S
I Year, B.E. Computer Science and Engineering
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1 November 2026

To
The Principal
St. Joseph's College of Engineering

OMR, Chennai – 600 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission.

Thank you for your kind consideration.

Yours obediently,
Harshitha S
(Register No.: [Register No.])

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a

AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

Question 30 (b) Personal email: Orientation programme

Selected Question: 30.b

To: [friend's email address]

Subject: My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Harshitha S

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.

4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.

   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.

   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound.

   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.

   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message.

   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.

2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically
   (b) flawlessly
2. (a) accurately

(b) instantly

3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Always wear insulated gloves and safety shoes while working.
2. Do not touch electrical equipment with wet hands.
3. Switch off the power supply before handling electrical connections.
4. Never touch exposed or damaged electrical wires.
5. Use properly insulated tools while working with machines.
6. Do not overload electrical circuits or equipment.
7. Keep a safe distance from rotating machines and moving parts.
8. Report electrical faults or accidents to the laboratory instructor immediately.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Always wear safety goggles, gloves, and protective clothing.
2. Do not operate CNC machines without proper training.
3. Keep your hands away from moving machine parts.
4. Secure the workpiece properly before starting the machine.
5. Never touch sharp tools or hot metal surfaces with bare hands.
6. Switch off the machine before making adjustments.
7. Keep the workshop clean and free from obstacles.
8. Report machine faults or accidents to the instructor immediately.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

A technical study tour to the Mettur Hydroelectric Dam is an interesting and educational experience for engineering students. Mettur Dam is located across the River Cauvery in Salem district, Tamil Nadu. It is one of the major dams in Tamil Nadu and is also known as Stanley Reservoir. The dam plays an important role in irrigation, water supply, and electricity generation.

During the study tour, students can observe the huge dam structure, water reservoir, power station, turbines, and generators. The engineers explain how electricity is generated using the power of flowing water. Water from the reservoir flows through large pipes called penstocks and rotates the turbines. The turbines are connected to generators, which convert mechanical energy into electrical energy.

The visit helps students understand renewable energy, hydroelectric power generation, water management, and modern engineering technology. Students can interact with engineers and gain practical knowledge about the working of a hydroelectric power station.

Students should follow safety rules during the visit. Always stay with the group, follow the instructions of the guides, avoid restricted areas, and keep the surroundings clean.

Overall, the technical study tour to Mettur Dam is a valuable learning experience. It connects theoretical knowledge with practical applications and motivates engineering students to learn more about science, technology, and renewable energy.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and

night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

A technical field visit to the Seawater Reverse Osmosis (RO) Desalination Plant in Nemmeli, Tamil Nadu, is an interesting and educational experience for engineering students. Nemmeli is located along the East Coast Road near Chennai. The desalination plant plays an important role in supplying drinking water to people by converting seawater into freshwater. It helps reduce water scarcity in Chennai and nearby areas.

During the visit, students can observe seawater intake systems, filtration units, high-pressure pumps, reverse osmosis membranes, and water storage tanks. First, seawater is collected from the sea and passed through filters to remove sand, dirt, and other impurities. Next, high-pressure pumps push the water through special semi-permeable membranes. These membranes separate dissolved salts and impurities from the water. Finally, the purified water undergoes further treatment before being supplied for public use.

Students should follow safety instructions during the visit. Always wear safety equipment, stay with the group, avoid touching machinery, and do not enter restricted areas. Overall, the technical field visit to the Nemmeli Desalination Plant provides valuable practical knowledge about water purification, automation, and modern engineering technology. It also teaches students the importance of water conservation and sustainable development.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Harshitha S
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Harshitha S
(On behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

Harshitha S
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Harshitha S

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Harshitha S
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Harshitha S
(On behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Harshitha S
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

I am writing this letter to request financial sponsorship and duty leave to participate in an upcoming technical event conducted at another institution.

The event provides an excellent opportunity to improve our technical knowledge, develop practical skills, and interact with students from other colleges. Participating in this event will help us improve our teamwork, communication, creativity, and problem-solving abilities.

We kindly request the management to provide financial support for registration fees, travel, and other necessary expenses. We also request duty leave for the participating students on [Date] to attend the event.We assure you that we will represent our college with discipline and bring recognition to our institution. We will also complete any missed academic work. Therefore, I kindly request you to consider our request and grant us necessary sponsorship and duty leave.

I would be grateful for your kind support and approval.

Thank you.
Yours obediently,
Harshitha S
(On behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV

produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Remote or virtual internships and on-site internships are two common ways for students to gain work experience and develop professional skills. Both provide opportunities to learn new technologies, improve communication skills, and understand workplace environments. However, they differ in their working methods, flexibility, and learning experiences.

Remote internships allow students to work from home using computers, the internet, and online communication tools. They provide flexibility, save travel time, and allow students to work with companies from different locations. Students can attend virtual meetings and complete tasks online. However, remote internships may provide fewer opportunities for face-to-face interaction and teamwork.

On-site internships require students to work directly at the company's office or workplace. Students can interact with experienced professionals, observe real-time operations, and learn through practical activities. They improve teamwork, communication, and problem-solving skills. However, they require travelling and following fixed working hours.

Both types of internships help students gain practical knowledge and prepare for their careers. Remote internships offer flexibility and convenience, whereas on-site internships provide direct interaction and hands-on experience.
In conclusion, both remote and on-site internships are valuable for students. Choosing the right type depends on individual needs, learning preferences, and career goals. A combination of both methods can provide a better professional learning experience.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Cloud computing infrastructure and on-premise data centers are two important methods used by organizations to store, manage, and process data. Both provide computing resources and help businesses manage their digital operations. However, they differ in terms of cost, accessibility, maintenance, and security.

Cloud computing uses internet-based services to store data and run applications. Companies can access resources from anywhere using an internet connection. It offers flexibility, scalability, and reduces the need to purchase expensive hardware. Popular cloud platforms include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud. However, cloud computing depends on internet connectivity and may involve ongoing subscription costs.

On-premise data centers are physical facilities where organizations maintain their own servers, storage devices, and networking equipment. They provide greater direct control over data, hardware, and security settings. They are useful for organizations that require complete control over their systems. However, they require high initial investment, regular maintenance, technical staff, and physical space.
Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws

carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

3D printing is an additive manufacturing process in which objects are created layer by layer using materials such as plastic, resin, or metal. It can produce complex shapes and customized designs with less material waste. It is useful for rapid prototyping and product development. However, 3D printing can be slower for large-scale production and may have limitations in strength and surface finish.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]

Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Harshitha S

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]

Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
Harshitha S

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]

Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

have a strong interest in becoming a AI Application and Agent Engineer in the future.

I have basic knowledge of C programming and good communication skills. I would like to improve my technical knowledge and gain practical experience. Since you have more experience, I would be grateful if you could guide me on which programming languages, tools, and online courses I should learn.

Please also suggest ways to improve my problem-solving skills and prepare for internships and placements. Your guidance will be very helpful in achieving my career goals.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving sister,

Harshitha S

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

Day 1: Overnight travel and arrival, check-in at the guest house and an orientation briefing. **Day 2:** Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. **Day 3:** Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** Morning group discussion, preparation of our tour report, and the return journey.

Safety protocols: Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

Estimated cost per student: travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

Day 1: Travel to the port city, check in and a short orientation. **Day 2:** Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. **Day 3:** Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. **Day 4:** Visit the control tower and the vessel traffic system, a feedback session and the return journey.

Dockyard safety protocols: Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

Category-wise expenditure per student: rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

Day 1: Overnight train journey and arrival. **Day 2:** Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. **Day 3:** Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. **Day 4:** Interaction with engineers, preparation of a tour report and the return journey.

Site safety rules: Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

Project Proposal: Ideation

Smart Canteen Seat Availability System | C Programming Prototype

1. Overview

Problem and Purpose

- The information should answer one practical question: which area is worth checking now, based on the latest recorded count.
- An estimate can support better decisions, but it cannot remove normal changes as students arrive, leave, or move tables.
- The proposal focuses on a small, visible campus problem that can be modeled and evaluated with beginner-level C programming.
- Students currently rely on visual checks or ask others whether seats are open; this can be unreliable when several areas are spread out.
- A shared estimate could reduce unnecessary walking and help staff understand which zones need attention during peak breaks.
- During short breaks, students may spend time searching for an empty table while seats remain available in other canteen areas. This can waste break time and create crowding near busy zones.
- The proposed system estimates available seats by zone and presents a clear count and status so students can decide where to look. Canteen staff could also review anonymous occupancy patterns.
- The first version is a console-based C program. It models zones and seating capacity, accepts sample or authorized counts, calculates estimated free seats, and labels each zone Available, Filling up, or Busy.

Scope and Expected Outcome

- The team will document assumptions about seat capacity, closed areas, and how often counts are refreshed.
- A short user walkthrough can confirm that the labels and instructions make sense to students unfamiliar with the project.
- If the walkthrough reveals confusion, revise wording before adding technical features.
- The prototype will use a small, clearly labeled sample dataset and make it easy to replace those records with authorized manual observations.
- A successful demonstration should let a student compare zones in a few steps and let a reviewer trace each displayed result back to its input.
- A later pilot may compare estimates with manual counts before any approved sensor, mobile app, or campus portal is considered.
- The system reports a zone-level estimate only. It does not identify people, reserve seats, or guarantee that a seat will remain free.
- The first milestone is a correct, understandable C prototype that can be demonstrated and tested using sample data.

Implementation and Review Notes

- Confirm the canteen layout with staff before defining zones, and note any seating areas that are temporarily unavailable.
- Use consistent zone names in menus, saved records, and demonstration material to avoid confusion.
- Keep an assumptions list so reviewers can distinguish agreed facts from prototype choices.
- Do not describe manually entered sample values as automatic or live occupancy readings.
- A brief setup guide should explain how to compile the program and load its sample data.
- Collect feedback on the wording before deciding whether the status labels need revision.
- The demonstration should state what the estimate can help with and what it cannot guarantee.

2. Goals, Scope and Success Criteria

A small, measurable first release

Goals

- Each menu action should have one clear result, such as viewing all zones, updating one count, or leaving the program.
- Use named constants for the status thresholds so the team can review and adjust the rule without searching through the program.
- Keep the menu short so a user can view the current summary or update one zone without navigating a complex workflow.
- Use the same calculation for single-zone details and the all-zone view to keep results consistent.
- Represent the canteen as named zones with confirmed seating capacities.
- Accept an occupied-seat count, validate it, and calculate free seats as capacity minus occupancy.
- Display a numeric estimate, a plain-language status, and the time of the latest update.
- Save and reload basic zone records using a simple text or CSV file.

Non-Goals

- Do not treat a displayed estimate as a reservation or direct students to enter an area that staff have closed.
- Online accounts, payments, and personal notifications are outside this learning prototype.
- The first release will not estimate future arrivals or decide where students should sit.
- The program will not collect names, student numbers, photos, or personal contact details.
- No individual identification, facial recognition, personal-device tracking, or attendance monitoring.
- No seat booking and no promise that a displayed seat will remain free.
- No live sensor hardware, machine-learning prediction, or mobile notifications in the first C version.

Success Criteria

- A new user should understand the meaning of Available, Filling up, Busy, Stale, and No recent count.
- The test notes should identify the compiler, sample file, inputs, expected results, and observed results.
- A reviewer should be able to reproduce the output from the documented input and verify the arithmetic manually.
- The program should explain when a result is sample, missing, or stale instead of presenting it as current occupancy.
- Reject negative counts and counts above the zone capacity without changing the last valid record.
- For valid input, show the correct difference and make all status labels understandable.
- Handle missing or malformed data without crashing and clearly identify sample data.

Implementation and Review Notes

- Make the first release achievable within the course timeline and available lab tools.
- Keep requirements measurable so the team can show whether each one has been met.
- A scope change should be recorded and reviewed for its effect on testing and completion time.
- Use a small sample file to make calculations easy to inspect during the presentation.
- Document why each excluded feature is deferred, especially hardware and online services.
- Success depends on correct results and clear communication, not on the number of features.
- Review the criteria with the instructor before treating the prototype as ready to demonstrate.
- A final review should compare the delivered functions with the original scope.
- Retest the application after any code or data-format change that could affect a requirement.
- Keep a short record of decisions so the team can explain why the first release is bounded.

3. Users and Requirements

Design around student and staff tasks

Users

- Students should be able to scan the summary in a few seconds and compare at least two zones.
- Staff should be able to correct an entry without needing to understand C or edit the source code.
- A student needs a quick comparison, not a detailed report; keep the most useful information visible together.
- Staff need a simple update process that fits around canteen operations and does not interrupt service.
- Students want to compare areas quickly and see whether a count is recent.
- Canteen staff may enter or review counts and understand busy periods.
- The student project team maintains the zone list, tests the program, and compares estimates with observations.

Core Functions

- A zone update should identify the previous count before replacement so accidental changes can be caught.
- The summary should remain useful when only some zones have recent counts; unknown values must be explicit.
- Provide a clear confirmation after an update and return to the summary so the user can see the changed estimate.
- Allow the user to cancel an update before saving if the wrong zone was selected.
- Add or load zones, record their capacities, and select a zone for an occupancy update.
- Validate each value, calculate free seats, and show one-zone detail or an all-zone summary.
- Save the latest count and update time, then load the record again on a later run.

Quality Requirements

- Use consistent terms throughout prompts, test cases, and documentation so the same status is not described differently.
- Keep source code readable with descriptive function names and comments for non-obvious rules.
- Accessibility: do not communicate status through color alone; use labels and readable contrast.
- Portability: avoid compiler-specific features where practical and document any assumptions made by the lab environment.
- Usability: short prompts, clear status words, readable output, and helpful error messages.
- Reliability: invalid records must not corrupt counts or crash the program.
- Privacy and transparency: store zone totals only, describe results as estimates, and show when they were updated.
- Maintainability: keep calculation rules separate from input, display, and file operations.

Implementation and Review Notes

- Use short prompts and familiar words so first-time users can follow the console flow.
- Organize functions around one responsibility to make errors easier to locate and correct.
- A user should be able to cancel before committing a new count to storage.
- Record staff feedback about the timing and practicality of entering counts.
- Test the summary with users who did not help build the interface.
- Keep accessible text labels in any future visual design, including for color-coded statuses.
- Use a consistent format for dates, times, capacities, and occupancy throughout the program.
- Avoid abbreviations in the menu unless their meaning is introduced and obvious.

4. C Technology and Data Model

A standard-library console prototype

Technology Choices

- Use a compiler and build command available in the college lab, and record their versions in the project notes.
- Keep dependencies minimal so the prototype can be compiled in a standard student lab environment.
- Use a documented text-file format that can be inspected and corrected without specialized software.
- Language: C using a compiler-supported standard agreed with the instructor; use standard headers such as stdio.h, stdlib.h, string.h, and time.h.
- Interface: a console menu and text-based zone summary. Data may be stored as delimited text with zone name, capacity, occupancy, and update time.
- Build and testing: document the college lab compiler and command, then run manual cases for calculations and file handling.

Program Components

- Keep file parsing separate from business rules so a future storage change does not alter the seat calculation.
- A menu controller coordinates the workflow, while small functions handle zone lookup, validation, calculation, and output.
- Return explicit success or failure results from file operations so the interface can report the outcome accurately.
- Zone management loads definitions and finds a selected zone. Input functions read and validate choices and counts.
- Availability logic calculates free seats and assigns the agreed status. Storage functions read and save records.
- Display functions show one zone or all zones, including the count source and update time when available.

Data Model

- Document a maximum number of zones and handle a full list with a clear message instead of writing past the array.
- Use one stable zone identifier or carefully validated name so updates cannot silently target the wrong area.
- Check the configured zone limit before adding a record and reject duplicate zone names to avoid ambiguous updates.
- If timestamps cannot be obtained reliably in the first demonstration, label them unavailable rather than inventing a time.
- Represent each zone with a bounded name, positive capacity, occupied count, and update time. A small fixed-size array is acceptable if its maximum size is checked.
- Keep zone definitions separate from changing occupancy values. Store only fields needed for the estimate; do not include names, IDs, images, or device identifiers.

Illustrative zone	Capacity	Occupied	Free estimate
Main Hall	80	62	18
Window Side	40	12	28
Upper Seating	50	50	0

The example values are illustrative, not measurements of the college canteen.

Implementation and Review Notes

- Write a small test for each core function before connecting all functions through the menu.
- Keep the selected C standard and compiler settings in the build instructions.
- Check file paths and permissions in the college lab environment before the final demonstration.

5. Availability Rules and Core Algorithm

Keep the estimate explainable and testable

Definitions and Statuses

- Treat a zone with zero usable capacity as unavailable or invalid, not as a percentage calculation.
- Keep the status wording neutral and avoid promising that a free seat will still be available on arrival.
- Calculate the percentage using a safe numeric type and round only for display; retain the original values for decisions.
- Agree on the threshold wording with students so “Filling up” is interpreted as a prompt to check, not a guarantee.
- Capacity is the number of usable seats in a zone. Occupied is the observed or entered count. Estimated free seats equals capacity minus occupied.
- A possible initial rule is Available when at least 40% of seats are free, Filling up when more than 0% but less than 40% are free, and Busy when none are free. Review thresholds with users.
- Always display the number and update time beside the label. If no reading exists, show No recent count; mark old values Stale according to an agreed time limit.

Calculation Steps

- Display whole-seat counts as integers and show a rounded percentage only as supporting information.
- Recalculate the summary after every accepted update so users do not see an old status beside a new count.
- Perform range validation before subtraction or percentage calculation to prevent impossible results and division by zero.
- Use one function for the status rule so a threshold change is applied consistently throughout the program.
- Load capacity and occupancy; reject non-positive capacity or occupancy outside zero through capacity.
- Calculate free seats, apply the agreed threshold, and display zone, estimate, status, and update time.
- Save only validated updates. Check calculations with empty, partly occupied, and full zones.

Examples

- Use a small zone in testing to check that percentage rounding does not incorrectly cross a status threshold.
- Include an unchanged update to verify that the program still reports the right zone and timestamp.
- Include boundary examples at the threshold and one seat below it to confirm the selected rule is applied consistently.
- Show a no-reading case separately so an unknown count is never confused with a zone that has zero occupants.
- With capacity 50 and occupancy 30, there are 20 free seats (40%), meeting the example Available threshold.
- With capacity 50 and occupancy 43, there are 7 free seats (14%), so the example status is Filling up. At 50 occupied, status is Busy.
- If occupancy is 52 for a capacity of 50, reject the value rather than displaying a negative estimate.

Implementation and Review Notes

- Keep the threshold calculation in one place and state its inclusive boundary explicitly.
- Use the same rounding rule in the display and in the test expectations.
- Retest all three status categories whenever a threshold is changed.
- Check that no calculation reads an uninitialized capacity or occupancy value.
- Include the latest update time whenever a status is shown to reduce ambiguity.
- Explain that an update may become outdated immediately as people move through the canteen.

6. Data Handling and Error Management

Protect count integrity and recover clearly

Data Fields

- Keep the file format documented with one example row and a description of each field.
- Trim surrounding whitespace from zone names and reject blank names before accepting a record.
- Use a consistent date-and-time representation so records remain easy to read and compare.
- Zone name: required, readable, and limited to a defined length.
- Capacity: positive integer within a documented prototype limit; occupancy: integer from zero through capacity.
- Update time: a readable timestamp or “unknown.” Record input source as manual or sensor only when known.

Input and File Safety

- Use a temporary output file or another safe write strategy so an interrupted save does not destroy the only valid data.
- Check every conversion result before using it; a partially parsed row should never be treated as a valid zone.
- Write updates only after every field has passed validation; do not leave a partly written record in the data file.
- Keep a backup of the last known-good sample file during development so tests can be repeated reliably.
- Read input into a bounded buffer, detect overly long lines, and check numeric conversion before using a value.
- When loading a file, verify required fields, separators, number conversion, and name length. Reject malformed records.
- Check that files open, writes succeed, and files close correctly. Report a save failure and retain the last valid record.
- Keep sample data clearly identified and never silently convert an invalid value into a plausible count.

Error Responses

- Include enough context in an error to identify the affected zone or file without exposing private information.
- Return to a safe menu option after recoverable errors and allow the user to retry or exit cleanly.
- After an error, preserve the menu state and tell the user the next safe action, such as retrying or reloading data.
- Use consistent messages for missing, invalid, and stale records so users can distinguish the conditions.
- Explain the accepted range for an invalid count; show configured choices for an unknown zone.
- If a file is missing or corrupt, explain whether sample data can be loaded or setup is needed. Do not imply the displayed sample is live.
- Separate parsing, validation, calculation, storage, and display into focused C functions so each can be tested.

Implementation and Review Notes

- Validate a complete record before changing the in-memory zone data.
- Handle end-of-file and unexpected input without repeating a prompt forever.
- Check save and close operations so a success message reflects a completed write.
- Keep a known-good sample file available for repeatable demonstrations.
- Review error messages with a user and remove jargon that does not help them recover.

7. User Interaction and Notification Concept

Make the estimate easy to read without implying a guarantee

Console Workflow

- Number menu options and repeat the valid range in an error message when the user enters an unsupported choice.
- After saving, confirm whether the record was written successfully and show the resulting estimate.
- List available zones before asking for an update and accept a clear menu choice instead of an ambiguous free-form command.
- Confirm the entered occupied count and zone before saving, especially when the update changes a prior value.
- Show a welcome message explaining that availability is an estimate, then offer choices to view zones, update a count, reload data, or exit.
- After an update, display zone name, free-seat estimate, status, and update time. If input is invalid, explain why and preserve the previous value.

Illustrative Output

Main Hall: 18 / 40 free (45%) | Available | updated 12:20 p.m.
Window Side: 5 / 24 free (21%) | Filling up | updated 12:19 p.m.
Outdoor Area: 0 / 16 free (0%) | Busy | updated 12:18 p.m.

The sample values are illustrative only, not live canteen measurements.

Future Display and Alerts

- A future screen should remain understandable with large text and keyboard navigation, not just a color-coded map.
- An alert should be suppressed if the underlying reading is stale or the system cannot confirm a recent update.
- Any future notification channel should be chosen with student consent and campus communication rules in mind.
- Keep the latest update time visible on any future display and provide a clear unavailable state if updates stop.
- Do not send an alert for every count change; notify only on a meaningful transition and provide a way to opt out.
- A future web or mobile page could show zone cards with status, numeric estimate, and “updated at” time. Use text as well as color and check readability and keyboard access.
- A notification could be opt-in and triggered by a meaningful status change. It should include a timestamp, avoid repeated alerts, and say that seats are not reserved.
- The C prototype can show a terminal message for a status change; it does not send SMS, WhatsApp, or push notifications.

Implementation and Review Notes

- Place the most recent information first and avoid decorative elements that compete with it.
- Show when a zone has no valid reading instead of suggesting it has no occupants.
- Allow users to return to the zone list after reviewing a single area.
- Any alert should be useful at the time it arrives and should not repeat unnecessarily.
- A future interface should be checked on both a phone-sized screen and a desktop.

8. Testing, Pilot and Evaluation

Build the logic first, then compare with observation

Test Cases

- Test the maximum allowed zone count and a list with one additional zone to verify capacity handling.
- Test names with spaces and punctuation, as well as blank and overlong names, to verify safe input handling.
- Repeat the same test after saving and reloading to verify that file handling does not alter a valid value.
- Test repeated menu use and cancellation to ensure a mistaken selection does not silently overwrite a count.
- Empty, full, and partial zones: verify the free-seat calculation and status.
- Negative and over-capacity counts: reject and retain the previous value. Zero capacity: reject and avoid division.
- Missing file, malformed record, long input, and failed save: show recovery guidance without crashing or claiming false success.

Pilot Plan

- Keep a consistent counting method during comparisons so differences are not caused by changing observation rules.
- Ask the canteen supervisor to identify temporary closures or maintenance that could affect the reference count.
- Choose observation periods that include both busy and quiet times, while avoiding disruption to normal service.
- Record whether an area is temporarily closed so its capacity is not treated as available seating.
- Select one zone with a clear boundary and confirm usable capacity with canteen staff.
- Use sample or manual counts first. If a sensor is considered, use it only after college approval.
- Compare system estimates with near-simultaneous manual counts during busy and quiet periods; record time, method, and unusual conditions.

Evaluation

- Report both average error and examples of large differences so a single summary does not hide poor cases.
- Use findings to decide whether the prototype is useful as a manual tool before considering a sensor.
- Summarize findings by zone and time period without recording the identities of people present.
- Use feedback to improve labels and workflow before considering a more automated counting method.
- Review absolute count error, status agreement, freshness of readings, coverage of planned times, and staff effort.
- Ask students whether they can identify a likely zone, notice the timestamp, and understand that a seat is not guaranteed.
- Set any numeric accuracy target after initial observations; report measured results and limitations honestly.

Development Stages

- Define zones and sample data; build and test C calculations; add console workflow and file handling; then consider an approved, bounded pilot.

Implementation and Review Notes

- Run the tests from a clean start so they do not depend on leftover data from a prior run.
- Repeat key tests with different zone capacities to catch assumptions tied to one example.
- Keep pilot observations separate from sample data and label both clearly.
- Record limitations alongside results so evaluation remains honest and reproducible.
- Use the pilot to learn whether updates can be maintained during real operating hours.
- End evaluation with a decision about whether to continue, revise, or stop the concept.
- Use repeated evidence to prioritize changes.

9. Risks, Limitations and Future Scope

Make uncertainty visible and respect privacy

Risks and Mitigations

- If users repeatedly see inaccurate estimates, pause the display and review the counting method before restoring it.
- Assign responsibility for refreshing data; otherwise even a correct program can show misleading information.
- A power or program interruption may leave data old; display its age and avoid implying that it is live.
- Crowding may move toward a zone shown as available; use a clear estimate disclaimer and review placement of any future display.
- Counts become outdated: show update time and label or hide stale readings.
- Manual entry errors: validate ranges, provide clear prompts, and compare with observations.
- Sensor misses or double-counts: test the device in the actual space and measure error before relying on it.
- Users expect a guarantee: describe the status as an estimate and do not reserve seats.
- Scope grows too large: finish and evaluate the C prototype before adding sensors or an app.

Limitations and Privacy

- A zone-level number cannot show whether seats are suitable for a particular group or accessible to a specific user.
- Avoid collecting or retaining individual movement data; the project goal can be met with aggregate counts.
- Manual counts can become outdated quickly, while automated counts may be inaccurate in crowded or obstructed spaces.
- Retain only aggregate information needed for evaluation and define who can access or remove it.
- The first version uses sample or manual input and cannot guarantee live occupancy. It estimates by zone rather than identifying individual empty chairs.
- Use anonymous zone-level totals only. Do not track students, use facial recognition, or collect personal device identifiers.
- Any sensor installation, campus display, or data-retention plan needs college review and clear ownership.

Future Scope

- Future additions should have a clear owner, maintenance plan, and measurable benefit before implementation.
- Accessibility review and staff feedback should accompany any move from a console prototype to a public interface.
- A future version could compare alternative counting methods, but only with approval and a transparent accuracy review.
- Consider integration with a campus portal only if a pilot shows clear student value and a team can maintain it.
- After testing, consider an approved counting method, mobile dashboard, aggregate peak-period summaries, or optional alerts.
- Each extension needs a review of accuracy, privacy, accessibility, reliability, and maintenance. Add only features supported by user need and pilot evidence.

Implementation and Review Notes

- Assign a person to confirm that a displayed count is recent before it is shared publicly.
- If input stops, show the last update time prominently or remove the stale estimate.
- Review any proposed equipment placement with staff to protect accessibility and clear walkways.
- Do not retain raw observations longer than needed for the agreed evaluation.
- Revisit privacy and accuracy before expanding to another canteen area.

10. Delivery Roadmap and Conclusion

Build in reviewable stages and expand only when evidence supports it

Roadmap and Deliverables

- Use small milestones with a review after each: requirements, working logic, file handling, testing, and final demonstration.
- Keep the sample dataset and test checklist with the source so another student can reproduce the output.
- Review the requirements with the instructor, then freeze the small first-release scope before coding begins.
- Keep a brief change log and identify which functions are complete, tested, or still planned.
- Define users, zones, assumptions, data fields, and limits; prepare illustrative records.
- Implement C data definitions, validation, calculation, console display, and file operations.
- Deliver source code, sample data, test checklist and results, and a short build/run guide.
- If approved, provide a pilot log and evaluation of error, freshness, effort, and user feedback.

Demonstration

- Explain the status threshold before showing results, and clearly state that all displayed counts are sample values.
- Finish with one limitation and one evidence-based next step to set realistic expectations.
- Use a short scenario with two zones so the audience can compare different occupancy levels and update freshness.
- Show one invalid value and explain how validation protects the displayed estimate.
- Introduce the campus problem, identify sample data, show the zone summary, update a valid count, and reject an impossible count.
- Save and reload the record, then explain the timestamp, privacy boundary, and limits. Distinguish implemented features from future ideas.

Conclusion and Pitch

- The first version prioritizes dependable calculations and clear communication over sensors or a polished interface.
- This staged approach gives the team a feasible C assignment and a responsible path to evaluate future improvements.
- The project is suitable for staged development because its core calculation is small, testable, and explainable in C.
- The next decision should depend on observed usefulness and accuracy, not on adding technology for its own sake.
- The Smart Canteen Seat Availability System is a practical CSE project: a C prototype can model zones, validate counts, calculate free-seat estimates, and display a status with its update time.
- A bounded pilot can show whether the estimate helps students and whether automation is worthwhile. The guiding principles are simplicity, transparency, privacy, and evaluation.
- Our goal is to help students choose where to look during short breaks. We will begin with a testable C program, compare its estimates with manual observations, and improve it from evidence before proposing a sensor or campus portal.

Implementation and Review Notes

- Prepare a concise handover so another student can compile, run, and test the program.
- Identify which deliverables are complete and which remain concepts for future work.
- Use audience questions to identify unclear assumptions, not as evidence of measured accuracy.
- Close the pitch by stating the next practical step: validate the prototype with authorized observations.
- Keep the project adaptable while protecting its small, testable first-release scope.

LinkedIn and GitHub Profiles

Part A: LinkedIn Profile

Harshitha S

CSE Student at St. Joseph's College of Engineering | Aspiring AI Application and Agent Engineer | Python, SQL, C (Basic)

About

I am a first-year Computer Science and Engineering student at St. Joseph's College of Engineering with a strong interest in artificial intelligence and software development. My long-term goal is to become an AI Application and Agent Engineer and build intelligent systems that address practical, real-world needs.

I am developing a foundation in programming and software engineering through Python and SQL, along with basic knowledge of C. I am interested in how AI models, data, tools, and user needs come together in useful applications and agents. I enjoy learning new concepts, working through challenging problems, and improving my understanding through practice.

Communication and leadership are strengths I bring to team settings. I value collaboration, thoughtful problem-solving, and continuous learning. As I progress through my degree, I plan to strengthen my computer science fundamentals and apply them through carefully documented projects.

Education and Current Experience

- B.E. Computer Science and Engineering, St. Joseph's College of Engineering | First year | 2026–2030.
- Student, St. Joseph's College of Engineering | August 2026–Present.

Current Skills

Area	Current foundation
Programming	Python; SQL; basic C programming.
Professional strengths	Communication, leadership, collaboration, and active learning.
Digital tools	Canva and Microsoft PowerPoint.
Career interests	AI applications, intelligent agents, software development, and practical problem-solving.

Current Focus

- Build confidence in programming fundamentals and problem-solving through regular practice.
- Learn data structures, algorithms, databases, and software development workflows.
- Explore AI and machine-learning concepts before building more advanced applications.
- Develop small projects, document the process, and share completed work responsibly.

Professional Direction

I am working toward a career that combines software engineering with practical AI. I want to understand how applications connect models to reliable data, how agents use tools safely, and how teams evaluate whether a system is useful. My immediate priority is building strong CSE fundamentals and turning new learning into small, explainable projects.

Links

Email: harshithasubii@gmail.com | LinkedIn: linkedin.com/in/harshitha-s-a6aa91439 | GitHub: github.com/harshithasubii0709

LinkedIn Profile Development

Part A continued: Focus, Projects and Career Roadmap

Areas of Interest

- AI application engineering: connecting models with useful data, software tools, and clear user workflows.
- AI agents: learning how applications can plan steps, use tools, and respond within defined limits.
- Software foundations: programming, databases, APIs, testing, and maintainable code.

Project Proposal: Smart Canteen Seat Availability System

This is a proposed C programming project, not a completed deployment. The concept is to estimate available seats in canteen zones and display a clear status with an update time. A first console prototype can use sample or authorized manual counts, validate capacity and occupancy, and calculate estimated free seats. Any sensor, dashboard, or notification feature would require testing and college approval.

- Learning goals: practice C structures, input validation, calculations, and file handling.
- Evaluation goal: compare estimates with manual observations in a small, approved pilot.
- Design boundary: show anonymous zone-level estimates; do not identify students or reserve seats.

Career Learning Roadmap

Stage	Learning focus	Evidence to build
Year 1	CSE fundamentals, Python, SQL, C basics, Git, and problem-solving.	Small exercises, clear notes, and one carefully documented beginner project.
Year 2	Data structures, APIs, databases, software testing, and introductory AI/ML.	Coursework projects with readable code, tests, and explanations.
Year 3	LLM application patterns, retrieval, evaluation, and agent tool use.	A practical AI application with documented design choices and limitations.
Year 4	System design, deployment, reliability, teamwork, and responsible AI.	A polished capstone and portfolio demonstrating end-to-end engineering.

Skills to Develop Over Time

- Strengthen C, Python, SQL, data structures, and algorithmic thinking through practice.
- Learn Git and GitHub workflows, APIs, testing, and collaborative development.
- Explore AI/ML, retrieval-augmented generation, model evaluation, and agent design as fundamentals grow.
- Add certificates or achievements only after they are completed and can be verified.

Professional Direction

My goal is to grow from a CSE student with programming foundations into an engineer who can design, build, evaluate, and communicate useful AI applications. I want my LinkedIn profile to reflect skills I can discuss today while making my learning direction clear.

Contact

Email: harshithasubii@gmail.com | LinkedIn: linkedin.com/in/harshitha-s-a6aa91439

Part B: GitHub Profile README

A student learning profile for GitHub | Username: harshithasubii0709

README

Hi, I'm Harshitha S, a first-year Computer Science and Engineering student at St. Joseph's College of Engineering.

I'm building a strong foundation in programming and software development, with a long-term goal of becoming an AI Application and Agent Engineer.

What I'm Learning

- Python and SQL; foundational C programming.
- Computer science fundamentals, data structures, and problem-solving.
- AI and machine-learning concepts, with an interest in practical applications and agents.

Current Project Direction

I am developing the idea for a Smart Canteen Seat Availability System as a beginner C project. The proposed console prototype would model canteen zones, validate occupancy counts, and show estimated free seats with a status and timestamp. It is a learning project concept; sensor input and a live dashboard are future possibilities that would need testing and approval.

How I Work

- I enjoy learning new concepts and improving through practice.
- I value clear communication, teamwork, leadership, and thoughtful problem-solving.
- I aim to make projects understandable with a clear purpose, setup instructions, examples, and known limitations.

As I complete projects, I plan to add their source code, a concise README, build and run steps, screenshots or demos when useful, and notes about what I learned. I will pin repositories only when they are ready for others to review and will keep private information and credentials out of public code.

Connect

Email: harshithasubii@gmail.com
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GitHub: github.com/harshithasubii0709

Projects to Document as They Progress

- C programming: the proposed canteen seat availability prototype, once a working version and tests are complete.
- Python and SQL: coursework or practice projects that clearly demonstrate concepts and include sample input and output.
- AI exploration: small experiments after the code, data sources, and limitations can be explained.

My GitHub profile is a place to document progress over time. Each repository should show what problem it explores, how to run it, what I contributed, and what I learned.